WORKED SOLUTION

STEP III 2013, Question 6

Pure • Complex Modulus Inequalities

STEP III 2013, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q6

This question has a full guided treatment in *Beyond STEP*, Section 5.8, *A Hidden Distance in the Disc*: https://beyondstep.keithmcnulty.org/ch05/poincare-disc.html

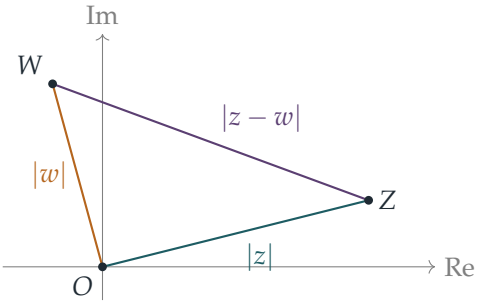
The key idea

Everything rests on writing a squared modulus as a number times its conjugate, $|u|^2 = uu^*$. Expanding $|z-w|^2$ and $|1-zw^*|^2$ this way produces the same cross terms $zw^* + z^*w$ in both, which is exactly why the single quantity E appears in each. Once both moduli are written as "something squared minus E", the final inequality is a comparison of two fractions of the form $\frac{A-E}{B-E}$ and $\frac{A}{B}$, and that comes down to the sign of $B - A$.

Solution

The triangle inequality

Mark the origin O and the points Z and W representing z and w in the Argand diagram. The distances OZ and OW are $|z|$ and $|w|$, and the distance WZ is $|z-w|$.



The straight segment WZ is the shortest path from W to Z, so it is no longer than the

path $W \to O \to Z$:

$$|z - w| = WZ \leq WO + OZ = |w| + |z|.$$

Equality holds exactly when O lies on the segment WZ, that is, when z and w point in opposite directions (or one of them is 0). If O, Z, W are collinear without O between them, the "triangle" is flat, but the shortest-path argument still applies.

Part (i)

Using $|u|^2 = uu^*$ and $(z-w)^* = z^* - w^*$,

$$\begin{aligned} |z-w|^2 &= (z-w)(z^*-w^*) \\ &= zz^* + ww^* - zw^* - z^*w \\ &= |z|^2 + |w|^2 - (zw^* + z^*w). \end{aligned}$$

Since $|zw| = |z||w|$, we have $(|z| + |w|)^2 = |z|^2 + |w|^2 + 2|zw|$, so

$$(|z| + |w|)^2 - E = |z|^2 + |w|^2 + 2|zw| - zw^* - z^*w - 2|zw| = |z-w|^2,$$

as required. Rearranging,

$$E = (|z| + |w|)^2 - |z-w|^2.$$

The right-hand side is a difference of squares of real numbers, so E is real. By the triangle inequality, $0 \leq |z-w| \leq |z| + |w|$, and squaring non-negative quantities preserves the inequality, so $E \geq 0$.

(Directly: z^*w is the conjugate of zw^*, so $E = 2\,\mathrm{Re}(zw^*) + 2|zw^*|$, which is real and at least 0 because $\mathrm{Re}\, u \geq -|u|$.)

Part (ii)

In the same way, using $(zw^*)^* = z^*w$,

$$\begin{aligned} |1 - zw^*|^2 &= (1 - zw^*)(1 - z^*w) \\ &= 1 - zw^* - z^*w + zz^*ww^* \\ &= 1 + |zw|^2 - (zw^* + z^*w) \\ &= \left(1 + 2|zw| + |zw|^2\right) - (zw^* + z^*w + 2|zw|) \\ &= (1 + |zw|)^2 - E, \end{aligned}$$

as required.

The final inequality

Write

$$A = (|z| + |w|)^2, \qquad B = (1 + |zw|)^2,$$

so that parts (i) and (ii) say $|z-w|^2 = A - E$ and $|1 - zw^*|^2 = B - E$.

First, the left-hand side makes sense: if $|z| > 1$ and $|w| > 1$ then $|zw^*| = |z||w| > 1$, so $zw^* \neq 1$ and $B - E = |1 - zw^*|^2 > 0$. Also $B > 0$.

Both sides of the required inequality are non-negative, so it is equivalent to the inequality between their squares,

$$\frac{A - E}{B - E} \leq \frac{A}{B}.$$

Multiplying through by the positive quantity $B(B - E)$, this is equivalent to

$$(A - E)B \leq A(B - E) \iff -EB \leq -EA \iff E(B - A) \geq 0.$$

Ask yourself What is the sign of $B - A$, and where does the condition on $|z|$ and $|w|$ come in?

Writing $a = |z|$ and $b = |w|$,

$$B - A = (1 + ab)^2 - (a + b)^2 = 1 + a^2b^2 - a^2 - b^2 = (1 - a^2)(1 - b^2).$$

If $a > 1$ and $b > 1$, both brackets are negative, so $B - A > 0$. Since $E \geq 0$ by part (i), $E(B - A) \geq 0$. Every step above was an equivalence, so

$$\frac{|z - w|}{|1 - zw^*|} \leq \frac{|z| + |w|}{1 + |zw|}.$$

Equality holds exactly when $E = 0$, that is, when z and w point in opposite directions.

When $|z| < 1$ and $|w| < 1$

Yes, the inequality still holds. Now $|zw^*| = ab < 1$, so again $zw^* \neq 1$ and $B - E > 0$, and the chain of equivalences goes through unchanged. This time both brackets in $(1 - a^2)(1 - b^2)$ are positive, so again $B - A > 0$, giving $E(B - A) \geq 0$ and hence the inequality.

What matters is only that $1 - a^2$ and $1 - b^2$ have the same sign. If one modulus is above 1 and the other below, $B - A < 0$ and the inequality can fail: for $z = 2$, $w = 0.4$ the left-hand side is $1.6/0.2 = 8$ while the right-hand side is $2.4/1.8 = \frac{4}{3}$.

Checks

- Take $z = 2$, $w = 2\mathrm{i}$. Then $|z - w| = 2\sqrt{2}$ and $1 - zw^* = 1 - 2(-2\mathrm{i}) = 1 + 4\mathrm{i}$, so the left-hand side is $2\sqrt{2}/\sqrt{17} \approx 0.686$, while the right-hand side is $4/5 = 0.8$. Scaling both by $\frac{1}{4}$ ($z = \frac{1}{2}$, $w = \frac{1}{2}\mathrm{i}$) gives the same two numbers, $0.686 \leq 0.8$, illustrating the $|z|, |w| < 1$ case.
- Equality case: $z = 2 + \mathrm{i}$, $w = -1.5z$ gives $E = 0$, and both sides equal 0.6577 to four decimal places.
- A computer check on several hundred thousand random pairs confirmed the two identities, that $E \geq 0$, and the inequality in both cases.

Common slips

- Writing $|z-w|^2 = |z|^2 + |w|^2 - 2zw^*$: the cross terms are $zw^* + z^*w$, which are conjugates of each other but not equal in general.
- Cross-multiplying without checking that $|1 - zw^*| \neq 0$; here $|zw| \neq 1$ guarantees it.
- Answering "no" to the last part because the derivation used $|z|, |w| > 1$: the factorisation $(1-a^2)(1-b^2)$ shows that only the agreement of signs matters.

Beyond STEP

Since $|1 - zw^*| = |1 - z^*w|$, the left-hand side is the modulus of $\dfrac{z-w}{1-z^*w}$. For fixed z with $|z| < 1$, the map $w \mapsto \dfrac{z-w}{1-z^*w}$ is a Möbius transformation taking the unit disc onto itself, and $\left|\frac{z-w}{1-z^*w}\right|$ is the pseudo-hyperbolic distance between z and w. The inequality in this question says that this distance is at most the "hyperbolic sum" of the distances from z and from w to 0: a strong form of the triangle inequality, and one of the foundations of hyperbolic geometry. Needham's *Visual Complex Analysis* explains Möbius transformations and the hyperbolic disc with beautiful pictures.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.